MATH 101 Unit 1: Expressions, Equations, and Inequalities

Practice 3 | Version 202610122020.36 | 30 minutes

Name: _____ Date: _____

Complete and show all work on separate paper. Write exact final answers at right. Basic four-function calculator only.

QUESTIONS	ANSWERS
1. Simplify. In (a), x and y are nonzero; give positive exponents. (a) $\frac{6x^{-4}y^{3}}{3xy^{-1}}$ (b) $\sqrt{125}$	1. (a) 10 pts (b) 10 pts
2. (a) Expand and simplify. (b) Factor completely. (a) $(4x-3)(x+4)$ (b) $2x^{2}-7x-15$	2. (a) 10 pts (b) 10 pts
3. Solve the inequality and write the solution in interval notation. $\lvert 2x-1\rvert \le 5$	3. Derived bounds 12 pts Interval 8 pts
4. Simplify and state all values excluded by the original expression. $\frac{x^{2}-16}{x^{2}+2x-24}$	4. Expression 14 pts Excluded values 6 pts
5. Solve both equations. In (b), state original restrictions and check your answer. (a) $x^{2}+2x-15=0$ (b) $\frac{6}{2x-8}+\frac{1}{x-4}=4$	5. (a) 10 pts (b), with check 10 pts

Instructor answer key: MATH 101 Unit 1: Expressions, Equations, and Inequalities

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Each column matches the vertical answer positions on its corresponding test page.

Concise grading reference; required work and checks are assessed on separate paper.

TEST PAGE 1

1. (a)

$\frac{2y^4}{x^5}$

10 pts

(b)

$5\sqrt{5}$

10 pts

2. (a)

$4x^2 + 13x - 12$

10 pts

(b)

$(x - 5)(2x + 3)$

10 pts

3. Derived bounds

$-2 \le x \le 3$

12 pts

Interval

$[-2, 3]$

8 pts

4. Expression

$\frac{x+4}{x+6}$

14 pts

Excluded values

$\{-6, 4\}$

6 pts

5. (a)

$x = -5,\ 3$

10 pts

(b), with check

$x = 5;\ x \neq 4$

10 pts